

Section B

2. Velothon Wales is a weekend of road cycling events. The Bloodwise cycling team raises funds to help beat blood cancer.

(a) Three cyclists are monitored before the Velothon. They have blood samples taken.

The results of their blood tests are shown in the table below. Normal ranges are also given.

Name of cyclist	Red blood cell count ($10^6/\text{cm}^3$)	White blood cell count ($10^3/\text{cm}^3$)	Platelet count ($10^3/\text{cm}^3$)
<i>Normal range</i>	<i>4.4 - 5.8</i>	<i>3.9 - 10.8</i>	<i>130 - 400</i>
Brian	5.5	35.1	240
Gareth	2.2	6.6	115
Jonathan	4.7	8.8	320

- (i) Suggest which cyclist needs treatment for an infection. [2]

Cyclist

Reason

- (ii) Suggest which cyclist needs treatment to raise their haemoglobin level. [2]

Cyclist

Reason

- (iii) Suggest which cyclist will experience a problem if he has a fall and cuts himself. [2]

Cyclist

Reason

- (b) Explain how the structure of veins and capillaries are related to their function. [4]

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- (c) (i) During a training ride, Jonathan accelerated from rest over a distance of 10 m in 2.5 s then travelled another 96 m at a constant speed in 6 s.
Use the equations:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration of the cyclist.

[4]

$$\text{Acceleration} = \dots\dots\dots \text{m/s}^2$$

- (ii) Jonathan has a height of 1.9 m. He is advised his maximum BMI should be 22.
Use the equation:

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

to calculate his maximum mass.

[2]

$$\text{Mass} = \dots\dots\dots \text{Kg}$$